

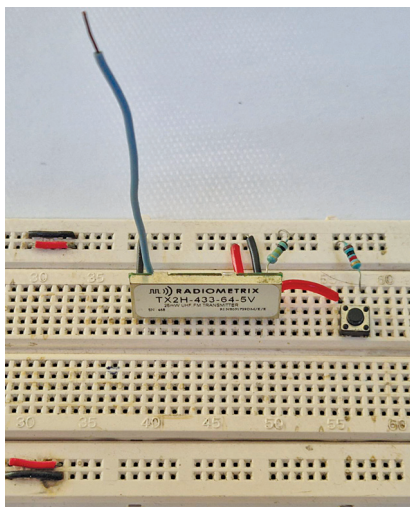


Fig. 3: Author's transmitter prototype

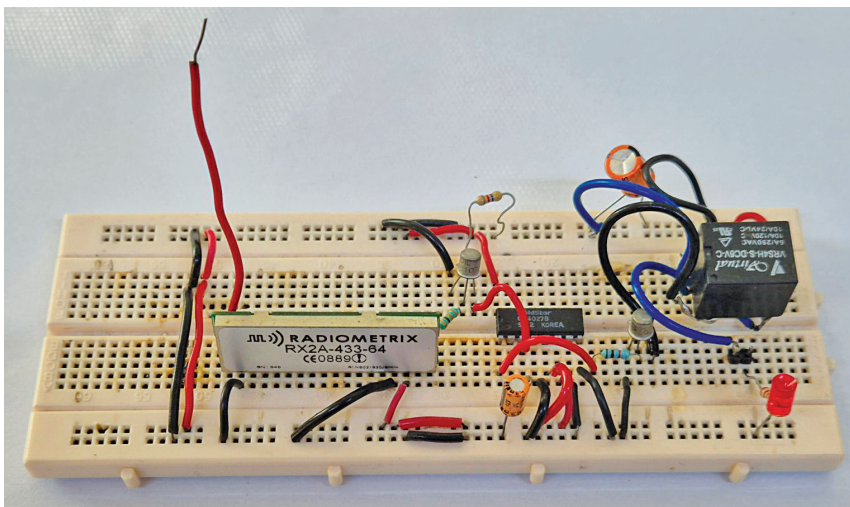


Fig. 4: Author's receiver prototype

ready to use. When LED1 is off, it means the relay is energised and the load is on. Capacitors C1 (220 μ F), and C2 (1 μ F) filter the power supply and suppress high-frequency noise, ensuring stable receiver performance. Resistors R3, R4, R5, and R6 provide biasing for the transistors and limit the current through LEDs and IC pins, thereby enhancing the circuit's stability.

When the transmitter sends an RF signal, the RX2A module demodulates it and passes the digital output to the CD4027 flip-flop via T1. Each valid signal toggles the flip-flop output, which drives T2 to energise or de-energise the relay. The relay contacts then switch the 230V AC load on or off.

It allows a single button to switch a device on or off alternately. The relay ensures isolation between low-voltage circuitry and the AC load for safety.

Construction and testing

Construction begins with assembling the transmitter on a breadboard or a general-purpose PCB. Mount the TX2H-433-64 module, push-button switch S1, and the required resistors, then solder a 17-centimetre antenna wire to the RF output. Place the completed transmitter inside a compact insulated enclosure.

The receiver should be assembled

on a separate breadboard or PCB. Install the RX2A-433-64 module, CD4027 IC, transistors, capacitors, resistors, relay, and flyback diode. Fit LED1 for power and status indication. Power the receiver using a regulated +5V DC adaptor. Connect the relay output to the 230V AC load through terminal blocks. Place the completed receiver unit in an insulated housing with terminals labelled appropriately.

After that, assemble the RX2A-433-64 module, CD4027 flip-flop IC, and driver transistors on another breadboard. Alternatively, you can construct on a general-purpose

PCB or a custom compact PCB for durability. Mount the 5V SPDT relay with a flyback diode for protection, and add the required capacitors and resistors for filtering, biasing, and noise suppression. Install LED1 to indicate system activity. Power the circuit with a regulated 5V DC adaptor, and connect the relay contacts to a 230V AC appliance (such as a lamp or fan) through terminal blocks (CON2 and CON3). Finally, house the assembly in an insulated plastic enclosure with clearly marked input and output terminals for safe use.

For testing, power the transmitter using a 5V battery and the receiver using a 5V adaptor. Connect a 60-watt lamp or similar appliance to the relay output. Press switch S1 to toggle the load on and off. Proper antenna positioning improves operational range. The system typically functions reliably up to 50 metres indoors and up to 100 metres in open areas. Suitable safety precautions must be taken when working with AC mains connections. **EFY**

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/live-diy-multi-use-rf-relay-control-system>

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PARTS LIST

Semiconductors:

IC1	- 4027 dual J K flip-flop
T1	- BC107 NPN transistor
T2	- 2N2222 NPN transistor
D1	- 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R5, R6	- 1k Ω
R2	- 100k Ω
R3	- 220 Ω
R4	- 4.7k Ω

Capacitors:

C1	- 220 μ F, 25V electrolytic
C2	- 1 μ F, 25V electrolytic

Miscellaneous:

CON1, CON2, CON3	- 2-pin connector
S1	- Push-to-on switch
RL1	- 5V single-pole, double-throw relay
ANT1, ANT2	- 17cm long wire
	- TX2H-433-64-5V transmitter
	- RX2A-433-64 receiver
	- Load (60-watt bulb)